TP2 Report: CNN and Transfer Learning for Age Estimation

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1. Part 1: A CNN from scratch

Test MAE and usefulness. The from-scratch CNN reached a test MAE of 13.62 years. A model that always predicts the mean training age (30.4 years) gets a test MAE of 14.39 years. The CNN is therefore only 0.77 years better than this trivial guess, which is too small to be useful in practice.

Overfitting or underfitting. At the best epoch (19), the train MAE is 11.08 years and the validation MAE is 11.67 years. The gap is small, so the model is not overfitting. Both errors stay high and the train loss only goes down slowly, from 426 to 209 in 20 epochs, so the model is underfitting. The validation curve is also unstable, for example it jumps to 15.27 at epoch 18. With about 4000 images and 20 epochs, the network cannot learn good face features from zero.

2. Experiment 1: Comparing the three approaches

Ranking. The fine-tuned ResNet is first with 8.79 years, the frozen ResNet is second with 12.38 years, and the from-scratch CNN is last with 13.62 years. This ranking matches what CM3 predicted, since starting from pretrained weights is better than learning from zero, and adapting these weights is even better.

Frozen vs fine-tuned. Unfreezing the backbone helped a lot, since the test MAE went from 12.38 to 8.79 years, an improvement of 3.59 years. However, the fine-tuning curves show strong overfitting. At the last epoch, the train loss is 10.15 while the validation loss is 99.34. In MAE, the model is off by only 1.71 years on the training images but by 7.36 years on the validation images. With 11 million parameters and only 4000 images, the network quickly memorises the training set. Fine-tuning a large pretrained network on a small dataset therefore works, but it needs a low learning rate and best-epoch selection to limit overfitting.

Why ImageNet features transfer to faces. The first layers of a CNN learn simple patterns like edges, colours and textures, which are useful for any natural photo. Age is visible through exactly these patterns, such as wrinkles, grey hair and skin texture. ImageNet also contains many photos of people. This explains why even the frozen ResNet beats the from-scratch CNN by 1.24 years.

3. Experiment 2: Does data augmentation help fine-tuning?

Effect on the test MAE. With augmentation, the test MAE is 8.30 years, against 8.79 years without augmentation. Augmentation therefore helped, but only by 0.49 years.

Train and validation gap. Without augmentation, the run ends with a train loss of 10.15 and a validation loss of 99.34. With augmentation, it ends with a train loss of 21.79 and a validation loss of 91.78. The train loss is higher because the model can no longer memorise the exact training images, and the validation loss is lower because the model generalises a bit better. The gap is therefore smaller but not closed, which is consistent with augmentation acting as a regulariser. The train MAE of the augmented run is measured on augmented images, so only the gap inside each run is compared.

Is the validation MAE a reliable signal? The best validation MAE is 7.04 with augmentation and 7.22 without, a difference of only 0.18 years. On the test set, the difference is larger, at 0.49 years, so the validation set almost hides the benefit of augmentation. With only 1200 validation images, a difference of a few tenths of a year can come from random noise. The validation score is also optimistic, because it is used to choose the best epoch. This is visible for every model, since the validation MAE is 1.3 to 1.9 years lower than the test MAE. The validation MAE alone is therefore not a reliable signal for such a small change, and several runs with different random seeds would be needed to confirm the effect.

4. Experiment 3: How much of the network should you fine-tune?

Position of partial fine-tuning. Partial fine-tuning reached a test MAE of 10.66 years, between the frozen model (12.38) and full fine-tuning (8.79).

Was the trade-off worth it? Partial fine-tuning trains 4,721,153 parameters (42.24%), against 513 for the frozen model and 11,177,025 for full fine-tuning. It improves on the frozen model by 1.72 years,

but it is still 1.87 years worse than full fine-tuning. It also overfits, with a train MAE of 2.24 against a validation MAE of 9.10 at the last epoch. The trade-off was therefore not really worth it here, because the earlier layers also needed to adapt to faces.

With only 500 training images. Full fine-tuning would look worse compared to partial fine-tuning. Even with 4000 images, full fine-tuning already memorises the training set, and with 500 images it would memorise the data even faster. Freezing more layers limits what the model can change, which protects it against overfitting.

5. Overall comparison

Method	Best epoch	Val MAE	Test MAE
ResNet (full FT + aug)	19	7.04	8.30
ResNet (full FT)	19	7.22	8.79
ResNet (partial FT)	20	9.10	10.66
ResNet (frozen)	19	10.50	12.38
From-scratch CNN	19	11.67	13.62

Ranking and size of the gains. The best model is the fully fine-tuned ResNet with augmentation, and the worst is the from-scratch CNN. Augmentation only adds 0.49 years on top of full fine-tuning, while going from the frozen model to full fine-tuning gives 3.59 years. The main gain therefore comes from fine-tuning, and augmentation only gives a small extra improvement.

Best epoch. All models reach their best epoch at 19 or 20, so they all used almost the full budget, and none of them peaked early and then got clearly worse. The fine-tuned models improve very fast in the first epochs and then only slowly. For example, full fine-tuning goes from 8.79 to 7.96 validation MAE in 3 epochs, and then only to 7.22 at epoch 19. The frozen model and the from-scratch CNN improve slowly during the whole run. A few epochs are therefore enough to get most of the gain of fine-tuning, while the other models would need more epochs.

Best accuracy for the training cost. The frozen model is the cheapest, but it is too weak for this task. Partial fine-tuning trains 42% of the parameters, but it loses almost 2 years compared to full fine-tuning. Full fine-tuning trains all parameters, but it gives by far the best accuracy, and augmentation adds no parameters and only a small extra cost. For a dataset of this size, full fine-tuning with augmentation is therefore the best trade-off.

6. Experiment 4: Feature map visualisation

The channels are numbered from 0, left to right, starting with the top row, and they clearly respond to different things. Channel 4 shows a blurred copy of the whole face, so it responds to bright skin areas and to the general face shape. Channel 5 lights up mainly along a curved vertical line on the right of the image, so it detects edges such as the contour of the face or the hair. Other channels, like 11 and 12, show small diagonal patterns spread over the image, so they respond to textures. These are low-level features, which is what an early layer like `layer1` is expected to learn.